



Stool-softening effect and action mechanism of free anthraquinones extracted from *Rheum palmatum* L. on water deficit-induced constipation in rats

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ABSTRACT

Ethnopharmacological relevance: In traditional Chinese herbal medicine, rhubarb is said to remove accumulation with purgation, clearing heat, and discharging fire. Modern pharmacology has shown that rhubarb extract has a purgative effect when given to experimental animals in an appropriate dose. However, the active components and their mechanism of action are still not clearly defined.

Aim of the study: The current research aimed to evaluate the synergistic stool-softening effects and explore the action mechanism of rhubarb free anthraquinones (RhA) and their monomers on constipation in rats.

Materials and methods: A rat model of water deficit-induced constipation was established to induce constipation, and these rats were treated with RhA and its monomers. ELISA, histopathology, immunohistochemistry, qPCR and Western blotting based on network pharmacology and molecular docking were conducted to explore the possible mechanism of action of RhA and its monomers.

Results: RhA, aloe-emodin, rhein, and chrysophanol showed stool-softening activity, and the combination of aloe-emodin and rhein had the strongest softening effect on faecal pellets. Aloe-emodin, rhein, and chrysophanol significantly increased the serum levels of vasoactive intestinal peptide (VIP), motilin (MTL), and substance P (SP), upregulated the expression of VIP, cyclase-associated protein 1 (CAP1), protein kinase A (PKA), cystic fibrosis transmembrane conductance regulator (CFTR), aquaporin 3 (AQP3), aquaporin 4 (AQP4), and aquaporin 8 (AQP8), decreased the expression of epithelial sodium channel (ENaC) and Na⁺/H⁺ exchanger 3 (NHE3), and reduced the colonic tissue concentration of Na⁺-K⁺-ATPase in the constipated rats. Osmolality of colonic fluid in model rats treated by RhA, aloe-emodin, rhein, and chrysophanol was increased.

Conclusion: Aloe-emodin, rhein, and chrysophanol were the stool-softening components of the RhA extract, and there were certain drug-interactions between the components. RhA upregulated VIP expression, activated the cyclic adenosine monophosphate protein kinase A (cAMP/PKA) pathway, and further stimulated CFTR expression while inhibiting NHE3 and ENaC expression, resulting in a hypertonic state in the colonic lumen. Water transport could then be driven by an osmotic gradient, which in turn led to the upregulation of AQP3, AQP4, and AQP8 expression. In addition, RhA likely improved gastrointestinal motility by increasing serum VIP, SP, and MTL concentrations, thus promoting faecal excretion.

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Abbreviations

Ae	Aloe-emodin	IOD	Integrated optical density
AOD	Average optical density	KEGG	kyoto encyclopedia of genes and genomes
AQP3	Aquaporin 3	MAPK	Mitogen activated protein kinase
AQP4	Aquaporin 4	Mod	Model group
AQP8	Aquaporin 8	MTL	Motilin
cAMP	CAP cyclic adenosine monophosphate cyclase-associated protein	Na ⁺ -K ⁺ -ATPase	Sodium-potassium pump
cAMP	Cyclic adenosine monophosphate	NHE3	Na ⁺ /H ⁺ exchanger 3
CFTR	Cystic fibrosis transmembrane conductance regulator	Nor	Normal group
Ch	Chrysophanol	PEG 4000	Polyoxyethylene 4000
ELISA	Enzyme-linked immunosorbent assay	Ph	Phycion
Em	Emodin	PI3K-Akt	Phosphatidylinositol-3-kinase
ENaC	Epithelium Na ⁺ channels	PKA	Protein kinase A
GO	Gene ontology	qPCR	Quantitative polymerase chain reaction
HPLC	High performance liquid chromatography	Rh	Rhein
IHC	Immunohistochemistry	RhA	Rhubarb free anthraquinones
		SP	Substance P
		VIP	Vasoactive intestinal peptide
		WB	Western blotting

1. Introduction

Constipation is a general digestive tract problem in clinical practice (Rao et al., 2016). Relevant research has shown that constipation is one of the risk factors for colonic perforation (Alsalmiy et al., 2018) and colon cancer (Rao et al., 2016; Verkuil et al., 2021). Laxative drugs are the most common treatments for constipation (Jiang et al., 2015), but long-term use may easily lead to drug dependence and cause gastrointestinal disorders and adverse reactions in the nervous system (Gibson et al., 2021). Therefore, it is vital to explore new drugs with fewer side effects.

Rhei Radix et Rhizoma, the medicinal part of rhubarb (*Rheum officinale* Baill., *Rheum palmatum* L., *Rheum Tanguticum* (Maxim. ex Regel) Balf.), is said to remove accumulation with purgation, clearing heat, cooling blood, and removing toxicity in traditional medicine (Pharmacopoeia Commission of People's Republic of China, 2020; Wang et al., 2022; Xiang et al., 2020). For example, Fan et al. reported that Rhei Radix et Rhizoma powder orally administered combined with abdominal massage helped acute stroke patients to prevent from constipation, achieved earlier health recovery, and improved quality of life (Fan and Li, 2008); Tian et al. reported that rhubarb can effectively prevent and treat patients with acute myocardial infarction accompanied by constipation (Tian et al., 2004). Modern pharmacological studies showed that rhubarb free anthraquinones (RhA) have a purgative effect when given to experimental animals in an appropriate dose (Liu et al., 2017b; Zhang et al., 2015). To date, there has been no systematic investigation of the ingredients that are responsible for relieving constipation, whether different active ingredients exhibit synergistic effects, and what their possible mechanisms of action are.

Network pharmacology and molecular docking have been widely applied for possible mechanism of action. Based on prediction results of network pharmacology and molecular docking, we hypothesized that the cAMP/PKA pathway and aquaporins (AQPs) may be involved in the occurrence of constipation. Recently, AQPs have been found to play an irreplaceable role in water transport (Sun et al., 2022; Suzuki, 2010). Aquaporin 3 (AQP3), aquaporin 4 (AQP4), and aquaporin 8 (AQP8) are highly expressed in the epithelial cells of the digestive tract, serving as channels for the transport and metabolism of biological fluids and electrolytes (Zhu et al., 2017). Vasoactive intestinal peptide (VIP) regulates gastrointestinal water transport by influencing AQP3 expression in its epithelial cells (Itoh et al., 2010). The expression of VIP can also influence the cyclic adenosine monophosphate protein kinase A (cAMP-PKA) pathway (Fernández et al., 2005), which in turn regulates the expression of cystic fibrosis transmembrane conductance regulator

(CFTR) and Na⁺/H⁺ exchanger 3 (NHE3) (Jiang et al., 2015; Kunzelmann and Mall, 2002). AQP-mediated water transport is also driven by osmotic pressure, which depends on the transport of transcellular ions (Ikarashi et al., 2016). VIP, cAMP, PKA, CFTR, and NHE3 proteins might be closely related to the expression of AQPs, thus balancing water transport between the basolateral side and the enteric cavity. Adenylate cyclase-associated protein (CAP) is the key enzyme in cAMP synthesis. After binding with the corresponding receptors on the cell membrane, hormone and other molecules can regulate the activity of CAP in the cytoplasm, and then regulate the content of cAMP in the cell to activate the related signaling pathway. Therefore, the level of CAP expression can be used to assess cAMP content. The CAP family mainly consists of CAP1 and CAP2. CAP1 shows a wide tissue distribution, while CAP2 is only significantly expressed in brain, heart, skeletal muscle, and skin (Ramsey et al., 2023). Therefore, the expression levels of CAP1 mRNA and protein were used to present the content of cAMP.

In this study, a water deficit-induced constipation rat model was adopted to individually investigate the stool-softening effects of RhA and its monomers. Furthermore, the synergistic effect of the different active monomers as purgatives was also studied. Histopathological observations, immunohistochemistry staining, qPCR, and Western blotting based on network pharmacology and molecular docking analyses were performed to explore the purgative mechanisms of RhA. *In situ* osmolality and concentrations of Na⁺ and Cl⁻ in colonic fluid were determined to support possible mechanisms. In addition, serum levels of VIP, motilin (MTL), and substance P (SP) and distal colonic level of Na⁺-K⁺-ATPase were also determined to elucidate the mechanisms of action involved.

2. Materials and methods**2.1. Materials**

Rhei Radix et Rhizoma, the root and rhizome of *Rheum palmatum* L. checked with MPNS (<http://mpns.kew.org>), was provided by Gansu Lixian Dahuang Sci & Tech Co. Limited, identified by Prof. Yanbin Shi (Lanzhou University). A voucher specimen (No. 830611) was deposited in the Faculty of Medicine of Lanzhou University. RhA monomers, including aloe-emodin (No.481-72-1), rhein (No.478-43-3), emodin (No.518-82-1), chrysophanol (No.481-74-3), and phycion (No.521-61-9), were purchased from Chengdu Push Biotechnology Co., Ltd. Polyoxyethylene 4000 was purchased from Chongqing Pharscin Pharmaceutical Co., Ltd. Rat VIP (E-EL-R1062c), SP (E-EL-0067c), and MTL (E-EL-R0639c), Na⁺ (E-BC-K207-M) and Cl⁻ (E-BC-K189-M) assay kits

were bought from Elabscience Biotechnology Co., Ltd. The Na⁺-K⁺-ATPase kit (A070-5-2) was purchased from Nanjing Jiancheng Bioengineering Institute. Trizol reagent was obtained from TaKaRa Bio Inc. HifairIII 1st Strand cDNA Synthesis SuperMix and HifairqPCR SYBR-Green Master Mix were bought from Yeasen Biotech Co., Ltd. (China). Antibodies including anti-VIP (ab8556), anti-NHE3 (ab95299), and anti-AQP4 (ab259318) were purchased from Abcam Co., Ltd. Anti-PKA (#4782) was purchased from Cell Signaling Technology, Inc. Anti-CAP1 (GTX112578) was purchased from GeneTex Co., Ltd. Anti-AQP3 (BA1559) was purchased from Boster Biological Technology Co., Ltd. Anti-ENaC (#DF9199) was bought from Affinity Biosciences Co., Ltd. Anti-β-actin (#21338), anti-AQP8 (#36812), anti-CFTR (#40734), and HRP-conjugated goat anti-rabbit Ig-G secondary antibody (#L3012) were bought from Signalway Antibody Co., Ltd.

2.2. Animals

Sprague-Dawley rats aged six weeks (male, 200 ± 20 g) were supplied by the Experimental Animal Center of Lanzhou University (License No. of Laboratory Animal Use Permit: SYXK, 2018-0002 [Gansu], Lanzhou, China). Animal experiments were approved by the Animal Experimentation Ethics Committee (20210301), School of Pharmacy, Lanzhou University and conducted in accordance with the European Community guidelines for laboratory animal use and care (86/609/EEC). All animals underwent adaptive rearing for 1 week before animal experiments.

2.3. Preparation of rhubarb free anthraquinones

A combined procedure for extraction and purification of rhubarb free anthraquinones has been reported by our research group (Wang et al., 2022). Briefly, 300 g of the crude powder of Rhei Radix et Rhizoma was soaked overnight in 70% ethanol with a ratio of powder to 70% ethanol of 1:8 (w/v) and then extracted twice by heat reflux at 70 °C for 2 h. The combined filtrates were evaporated under reduced pressure to recover ethanol and then extracted twice by dichloromethane (1:1, v/v). The organic extract was combined, and the dichloromethane was evaporated and recovered. The brown residue was dried in the dark to obtain the rhubarb free anthraquinones, and the yield of the RhA extract (%) was obtained by dividing the mass of the RhA extract by the mass of crude powder (300 g).

2.4. Experimental design

2.4.1. Establishment of dry-stagnation type constipation

The rats were randomly separated into the normal control group (Nor) and constipation model group (Mod). Referring to the modeling method of dry-stagnation type constipation recorded in the book *Experimental Methodology of Pharmacology* (Wei, et al., 2010) and based on the pilot studies, a constipation model of rats was induced by limiting water intake, but the rats had free access to food. Briefly, the rats were permitted free access to standard food but only 3.0 mL water intake twice daily (12 h interval) for 5 days. The weights of the rats were recorded on each day. Faeces were collected daily to determine faecal water content, and the number and weight of faecal pellets within 6 h were recorded.

2.4.2. Stool-softening experiment with RhA

The constipation model rats were separated into five groups with six rats per group: a model control group given vehicle alone (Mod), a positive control group given polyoxyethylene 4000 (PEG 4000, 1800 mg/kg), a low dose RhA group (LRhA, 100 mg/kg), a medium dose RhA group (MRhA, 300 mg/kg), and a high dose RhA group (HRhA, 500 mg/kg). The treatments started on the day five of rat modeling and lasted for 3 consecutive days. The samples for oral administration by gavage were prepared with the Tween-80 (2%, w/v) and carboxy methylcellulose

sodium (CMC-Na) (0.5%, w/v) as vehicle which osmolality was about 35 mosm/kg H₂O. The normal control group (Nor) with six rats was given an equal volume of normal saline. The rats in the Nor group fed and drank freely, and the rats in the other groups fed freely but their water intake was limited until the end of the experiment.

2.4.3. Stool-softening experiment with RhA monomers

The constipation model rats were separated into seven groups with six rats per group: a model control group given vehicle (Mod) and RhA treatment (RhA, 300 mg/kg), aloe-emodin treatment (Ae, 300 mg/kg), rhein treatment (Rh, 300 mg/kg), emodin treatment (Em, 300 mg/kg), chrysophanol treatment (Ch, 300 mg/kg), and physcion treatment groups (Ph, 300 mg/kg). The normal control group (Nor) was given an equal dose of normal saline. The treatments were given for 3 consecutive days.

2.4.4. Stool-softening experiment with combinations of RhA monomers

The effective anthraquinone monomers were mixed in combination according to the mass ratio of each monomer in RhA. The average percentage (w/w, %) of Ae, Rh, Em, Ch and Ph was separately 10.84%, 20.64%, 23.16%, 30.98%, 8.94% (Supplementary Table S1) (Wang et al., 2022). In order to facilitate weighing and sampling, the mass ratio of Ae, Rh, Em, Ch and Ph was roughly set as 1:2:2:3:1. The experimental groups were the aloe-emodin, rhein, and chrysophanol group (Ae + Rh + Ch, 1:2:3, w/w/w), the aloe-emodin and rhein group (Ae + Rh, 1:2, w/w), the aloe-emodin and chrysophanol group (Ae + Ch, 1:3, w/w), and the rhein and chrysophanol group (Rh + Ch, 2:3, w/w). Additionally, the RhA treatment (RhA), model control (Mod), and normal control groups (Nor) were arranged in parallel. The tested samples were gavaged at 300 mg/kg for 3 consecutive days.

2.5. Evaluation of stool-softening effect of RhA and its monomers

2.5.1. Measurement of faecal water content

Fresh faeces of individual rats were collected, weighed, dried at 70 °C for 24 h, and re-weighed. The faecal water content was calculated according to the following formula:

$$\text{Faecal water content (\%)} = \frac{\text{fresh faeces weight} - \text{dry faeces weight}}{\text{fresh faeces weight}} \times 100\%.$$

2.5.2. Measurement of the number and weight of faecal pellets

Each rat was observed in a metabolic cage. Faeces from each rat were collected and blotted on white filter paper, and the total number and wet weight of faecal pellets of each rat over 6 h were recorded.

2.5.3. Colonic bead expulsion test

Colonic propulsion was evaluated applying the bead expulsion test 12 h from the last treatment. After the rats were anesthetized by ethyl ether, 5.5 mm beads were inserted into the distal colon about 3 cm from the anus. Bead expulsion latency from bead insertion to expulsion was recorded (Zhao et al., 2022).

2.5.4. Intestinal carbon powder propulsion test

After the last treatment, the rats were fasted overnight; then the intestinal propulsion experiment was performed to determine gastrointestinal motility function. Carbon powder dispersed in normal saline (10%, w/v) was fed to the rats at a dose of 10 mL/kg. After 30 min, the rats were rapidly euthanized by chloral hydrate, and the small intestine was gently peeled off and spread out on filter paper. The intestinal propulsion rate was obtained by measuring the total length of the small intestine (L1) and the length of the small intestine containing semisolid paste (L2), calculated using the formula $L2/L1 \times 100\%$. Additionally, colonic tissue was obtained for histopathology, immunohistochemistry, gene and protein analyses.

2.6. Biochemical indices in serum and colonic tissues

At the end of the treatment, rats were anesthetized by diethyl ether prior to cardiac puncture to collect blood and centrifuged at 4000 rpm for 30 min to obtain serum. The levels of VIP, MTL, and SP were determined using ELISA kits according to the manufacturer's instructions, while Na⁺-K⁺-ATPase was measured using commercially available kits as per manufacturers' instructions, and the results were presented as μmolPi/mgprot/h.

2.7. Network pharmacology and molecular docking

2.7.1. Prediction of potential targets of rhubarb against constipation

Using the network pharmacology approach, the multiple components, critical targets and potential mechanisms of rhubarb against constipation were investigated. "Rhubarb" and "constipation" were used as keywords to search the relevant databases. The compounds of rhubarb were obtained from the databases of TCMSP and TCMIP. These compounds and their potential targets were screened through the PubChem database, Swiss Target Prediction, ETCM and Tasly-TCM databases. Meanwhile, the targets of constipation were collected by searching the OMIM, the MalaCards and the GeneCards database. The overlapping targets of rhubarb and constipation were obtained using Draw Venn Diagram. STRING database was adopted to obtain core targets. To explore the biological function of potential targets in constipation, GO enrichment and KEGG pathway analysis were conducted.

2.7.2. Molecular docking analysis

The RCSB PDB database and PubChem database were used to download the stereo structures of core targets and active ingredients. They were preprocessed by PyMol software, and then imported into AutoDock software to calculate the binding energy between the proteins and small molecule ligands.

2.8. Histopathologic examination and immunohistochemistry

Colonic tissue was rinsed with normal saline and fixed with a 4% paraformaldehyde solution for 24–48 h. The fixed tissue was embedded in solid paraffin and cut into 5-μm-thick slices. The slices were stained with hematoxylin and eosin (H&E) and examined under a microscope.

The paraffin-embedded sections were stained with an avidin-biotin complex (Li et al., 2019), incubated with the primary antibodies, including anti-NHE3 antibody (1:25), anti-ENaC antibody (1:100), anti-AQP3 antibody (1:100), and anti-AQP8 rabbit polyclonal antibodies (1:100), overnight at 4 °C. Subsequently, each slide was incubated with biotinylated anti-rabbit IgG secondary antibodies and then with avidin-biotin-horseradish peroxidase complex, followed by staining with DAB and Mayer's hematoxylin solution for 5 min, and rinsed with water for 15 min. The above tissues were dehydrated sequentially with 75–100% ethanol, immersed in xylene, and sealed with resin. Microscopic observation was performed using an Olympus BX60 optical microscope (Tokyo, Japan). The brownish yellow particles in the image were the target proteins, which served as the positive standard to analyze all photographs using the Image-Pro package. The package was also used to estimate the integrated optical density (IOD) and tissue area of each photograph, and then, the average optical density (AOD) value was calculated according to the following formula:

AOD = $\frac{\text{IOD}}{\text{tissue area}}$

The larger values of AOD suggested higher expression levels of the target proteins. The positive immunostaining slide was analysed at three random visual fields at a magnification of × 200.

2.9. Quantitative RT-PCR analysis

Total RNA from the colonic tissue was extracted using Trizol reagent as per the manufacturer's instructions. The absorption ratio of total RNA from 260 nm to 280 nm was determined using a NanoDrop ONE instrument (Thermo Fisher Scientific Inc., USA) to check the purity of RNA (Lu et al., 2015). Possible adulterating DNA was removed, and the RNA was reverse transcribed into a cDNA template. A HifairIII 1st Strand cDNA Synthesis SuperMix was used for quantitative analysis. The cDNA products were separately amplified by polymerase chain reaction (PCR) with specific primers for VIP, cAMP, PKA, CFTR, NHE3, epithelial sodium channel (ENaC), AQP3, AQP4, AQP8, and β-actin (Table 1). The expression levels of the target genes were analysed with a QuantStudio 3 real-time PCR system (Thermo Fisher Scientific Inc., USA). mRNA abundance was quantified by the 2^{-ΔΔCt} value method. The values were adjusted by β-actin quantitation and expressed as a gene/β-actin ratio.

2.10. Western blotting analysis

About 30 mg of distal colon was isolated and transferred into 1 ml of frozen RIPA lysis buffer and 100 mmol/L of PMSF. Tissues were homogenised and centrifuged. The supernatant was collected to determine the total protein concentration using a BCA protein assay kit. The samples were diluted in SDS-PAGE loading buffer at 100 °C for 5 min and subjected to SDS-PAGE. After transfer to a polyvinylidene difluoride (PVDF) membrane, western blotting membranes were stripped, blocked, and incubated overnight at 4 °C with VIP (1:100), CAP1 (1:3000), PKA (1:1000), CFTR (1:1000), NHE3 (1:1000), ENaC (1:1000), AQP3 (1:1250), AQP4 (1:1000), AQP8 (1:1000), and anti-β-actin (1:5000) antibodies. After washing with TBST, the membranes were incubated with HRP-conjugated goat anti-rabbit Ig-G secondary antibody for 1 h at ambient temperature and then visualised using the super ECL detection reagent. Finally, the target protein bands were scanned using FUSION SOLO6S.EDGE (VILBER, France), and the gray scale was quantitated using Image J.

Protein level = $\frac{\text{the gray scale of each protein}}{\text{the gray scale of } \beta\text{-actin}}$

2.11. Osmolality and electrolyte ions in colonic fluid

The constipation model rats were separated into seven groups, fasted for 16 h before experiment. The colon was exposed and ligated at both ends after the rats were anesthetized by inhalation of ethyl ether. The

Table 1
Primer sequences for quantitative real-time PCR assays.

Genes	Gene bank	Primer sequences
β-actin	NM'031144	Forward:5'-CTGTGTGGATTGGTGGCTCT-3' Reverse: 5'- GCAGCTCAGTAACAGTCCGC -3'
VIP	NP'446443	Forward:5'-AATCCAGAAGCAAGCCTCAGTTCC-3' Reverse: 5'-GCAGCCTGTCATCCAACCTCAC-3'
CAP1	NP'071778	Forward:5'-TGGAGGCGGTGTACATACTTC-3' Reverse: 5'-GGATTGGCAAGCAGCGAGTC-3'
PKA	NM'001100922	Forward:5'-TTGGACAAGCAGAAGGTGGTGAAG-3' Reverse: 5'-CCAGGAACGGGAAGTTGACAGC-3'
CFTR	NM'031506	Forward:5'-TTGTTGGGAATCAGCGATGGAGAAG-3' Reverse: 5'-GAAGAAGGCAGAGCTAGTGAAGAACC-3'
NHE3	NM'012654	Forward:5'-CTGGCGTGGACTGTGTGAAAGG-3' Reverse: 5'-GACAGCAGGAAGGCGAAGATGAC-3'
ENaC	NM'031548	Forward:5'-CACCGTCTCTGTCTCCATCAAAGTC-3' Reverse: 5'-CCAAGTCAGTCAGAAGTCACTCAC-3'
AQP3	NM'031703	Forward:5'-CTGTGGTTCGGTGGCTCAAGTG-3' Reverse: 5'-GATGGCAAGGGTGACAGCGAAG-3'
AQP4	NM'001142366	Forward:5'-GTGATTCCAACGGACTGATG-3' Reverse: 5'-TTGGTCTTTCCCTTCTTCTC-3'
AQP8	NM'019158	Forward:5'-TGTGTGTATGGGTGCCGTCAATG-3' Reverse: 5'-GCCATCACAGCAGGTCCAAAGG-3'

colonic lumen was rinsed with normal saline at 37 °C. Then the colonic lumen of model control group (Mod) was injected 2 mL of Kreb's solution (Kreb's: g/L, NaCl 7.05, KCl 0.44, NaH₂PO₄·2H₂O 0.19, MgCl₂·6H₂O 0.24, NaHCO₃ 1.29, C₆H₁₂O₆·H₂O 2.28, CaCl₂ 0.28), and RhA, Ae, Rh, Em, Ch and Ph treatment groups were separately given corresponding drugs dissolved in Kreb's solution at a concentration of 15 mg/mL, and each rat was given 2 mL. The normal group (Nor) was given an equal dose of Kreb's solution. The rats were sacrificed by intraperitoneal injection of chloral hydrate 1 h after treatment, and the osmolality of colonic fluid was determined by a freezing osmometer (YASN Lab Technologies, UK). The Na⁺ and Cl⁻ were measured using commercially available kits as per manufacturers' instructions, and the results were presented as mmol/L.

2.12. Statistical analysis

The data were expressed as mean ± standard deviation (SD) and statistically analysed by one-way analysis of variance (ANOVA) if the data were normally distributed and had homogeneous variance, and the LSD method was used for multiple comparison. A *P*-value <0.05 was considered statistically significant, while *P* < 0.01 was considered highly significant. The image analyses were performed using GraphPad Prism v8.3.0 and/or Image J software.

3. Results

3.1. Chemical constituents of RhA

In our previous study, the identification of the chemical constituents of RhA and the quantitative analysis of RhA have been reported (Wang et al., 2022). Ae, Rh, Em, Ch, and Ph were the main chemical constituents of RhA (Fig. 1), and the identification data are shown in the Supplementary materials (Suppl Figs. S1, S2 and S3). The yield of the RhA extract was approximately 0.98% (w/w). The percentage of the five free anthraquinones (Fig. 1) in the RhA extract was more than 92.0% (w/w), and the mass ratio of Ae, Rh, Em, Ch, and Ph was approximately 1:2:2:3:1 (Suppl. Table S1).

3.2. Stool-softening effect of RhA on rat constipation

3.2.1. Verification of dry-stagnation type constipation rat model

After water intake was restricted in rats, food intake decreased and hypoactivity, depilation, and messy hair were observed. On the second day of modeling, their body weights began to decrease (Fig. 2A). The faeces decreased from the second day on, gradually shriveled, and

became dry and sticky on the 5th day. The urine became reddish yellow, and the anus became dirty. The tips of their noses were blood red and swollen, and their tongues were purplish red. These signs met the criteria of the dry-stagnation constipation model (Chen et al., 2019).

Faecal water content, faecal pellet number, and faecal weight were determined to assess defaecation function. As illustrated in Fig. 2B–D, compared with the Nor group, the defaecation function of the Mod group was significantly impaired from the third day on. Taken together, these data confirmed that the constipation rat model was successfully established.

3.2.2. Stool-softening effect of RhA

After 3 days of experimental drug intervention, compared with the Mod group, in the LRhA group, there were no significant differences in faecal water content and in the number and weight of faecal pellets within 6 h, while they were significantly increased in the MRhA and HRhA groups (*P* < 0.01) (Fig. 3A–C). Similarly, the colonic bead expulsion time of the MRhA and HRhA groups was significantly shortened (*P* < 0.01) (Fig. 3D), and the small intestinal propulsion rate of the MRhA and HRhA groups was significantly increased compared with the Mod group (*P* < 0.01) (Fig. 3E). All indices of the PEG 4000 group showed no significant differences from the MRhA and HRhA groups, and they were closer to the Nor group. These results suggested that the test dose of RhA for the treatment of rat constipation could be set at 300 mg/kg per day for 3 consecutive days. As shown in Fig. 3A–E, all evaluation indicators of the LRhA group versus MRhA or HRhA group showed a dose-dependent manner (*P* < 0.05).

3.2.3. Stool-softening effect of RhA monomers

As shown in Fig. 4A, after 3 days of experimental drug intervention, the faecal water contents of the RhA, Ae, Rh, and Ch groups were significantly greater than that of the Mod group (*P* < 0.05). The number and weight of faecal pellets within 6 h of the Mod group were significantly reduced compared to the Nor group, while more faecal pellets over 6 h were found in the RhA, Ae, Rh, and Ch-treated groups compared with the Mod group (*P* < 0.01) (Fig. 4B and C). As shown in Fig. 4D, colonic bead expulsion latency of the Mod group was significantly prolonged compared with that of the Nor group, but it was shortened after treatment with RhA, Ae, Rh, Em, and Ch (*P* < 0.05). The small intestinal propulsion rate of the Mod group was significantly decreased compared to that of the Nor group, but it was increased after treatment with RhA, Ae, Rh, and Ch (*P* < 0.01) (Fig. 4E). Taken together, the results demonstrated that RhA, Ae, Rh, and Ch possessed excellent stool-softening effects by improving intestinal and colonic motility.

3.2.4. Stool-softening effect of combinations of RhA monomers

After 3 days of experimental drug intervention, the faecal water contents of the Ae + Rh + Ch, Ae + Rh, Ae + Ch, and Rh + Ch groups were significantly increased in comparison with the Mod group (*P* < 0.01) (Fig. 5A); the 6-h defaecation number, weight, and small-intestinal propulsion rate were all significantly increased in the Ae + Rh + Ch group, Ae + Rh group, Ae + Ch group, and Rh + Ch group compared to the Mod group (*P* < 0.01) (Fig. 5B, C, and 5E). Noticeably, the faecal water content, 6-h defaecation number, and weight of the Ae + Rh group were all higher than those of the other treated groups (*P* < 0.05) (Fig. 5A–C), while the colon expulsion latency of the Ae + Rh group was significantly shorter than that of the other treated groups (*P* < 0.05) (Fig. 5D). In summary, the compatibility of RhA monomers among Ae, Rh, and Ch demonstrated a definite stool-softening effect, and the combination of Ae and Rh showed the greatest stool-softening effect.

3.3. Levels of biochemical indices in serum and colonic tissues

As shown in Fig. 6A1–A3, the levels of serum VIP, SP, and MTL of the Mod group were significantly lower than those of the Nor group. After treatment with RhA, Ae, and Ch, the levels of VIP, SP, and MTL

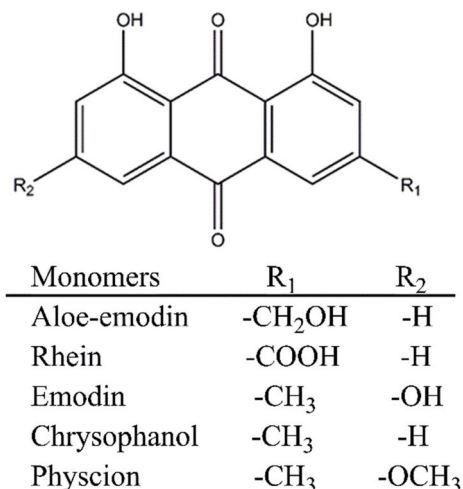


Fig. 1. Chemical structure of five free anthraquinones.

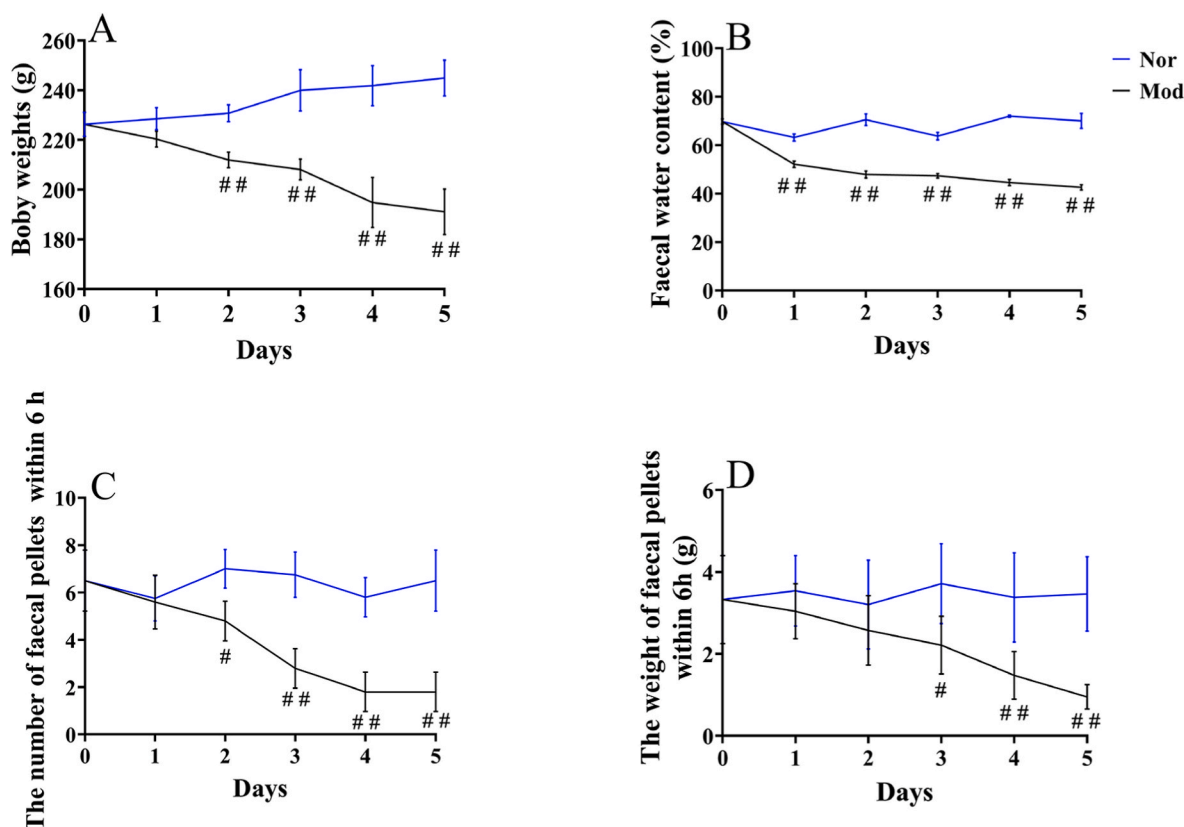


Fig. 2. Changes in defaecation function in the model rats. (A) Weight changes; (B) faecal water content; (C) number of faecal pellets; (D) weight of faecal pellets. # $P < 0.05$, ## $P < 0.01$ vs. Nor group.

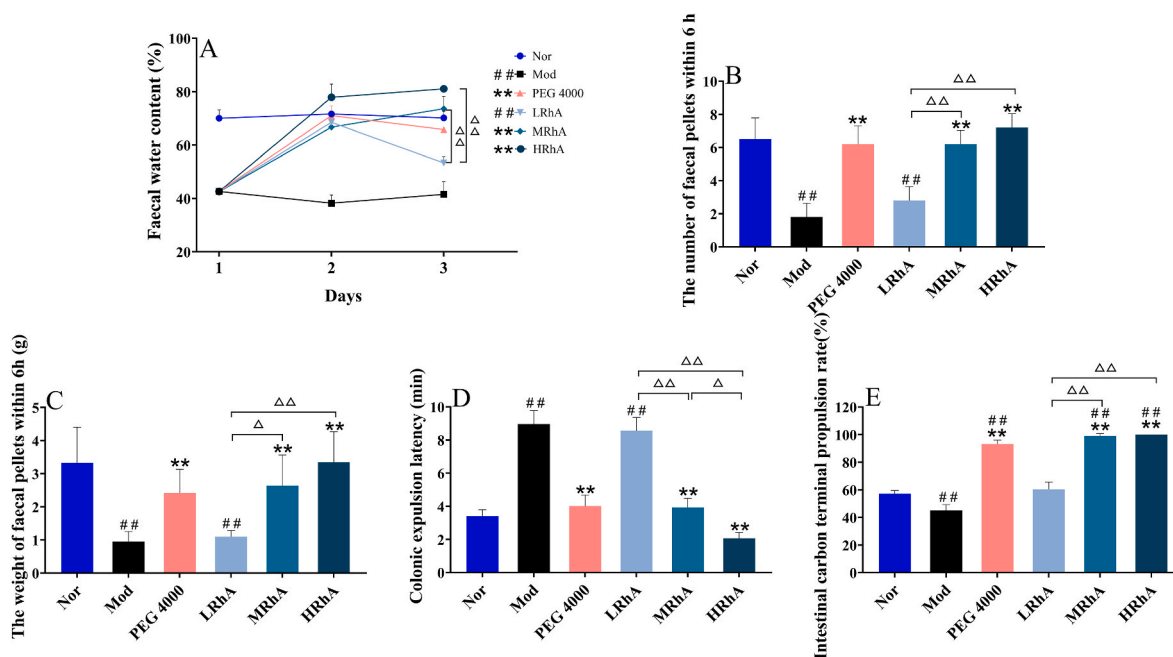


Fig. 3. Effects of RhA and PEG4000 on stool-softening. (A) Faecal water content; (B) number of faecal pellets; (C) weight of faecal pellets within 6 h; (D) colonic expulsion latency; (E) small intestinal propulsion rate. # $P < 0.05$, ## $P < 0.01$ vs. Nor group; * $P < 0.05$, ** $P < 0.01$ vs. Mod group, $\Delta P < 0.05$, $\Delta\Delta P < 0.01$ vs. between LRhA, MRhA and HRhA.

recovered to normal values ($P < 0.05$). Surprisingly, the level of MTL of the Rh group was significantly lower than that of the Mod group ($P < 0.01$) (Fig. 6A3). Na^+/K^+ -ATPase in the colonic tissues of the Mod group was significantly higher than that of the Nor group, while all treatments

significantly decreased Na^+/K^+ -ATPase to normal levels ($P < 0.01$) (Fig. 6B).

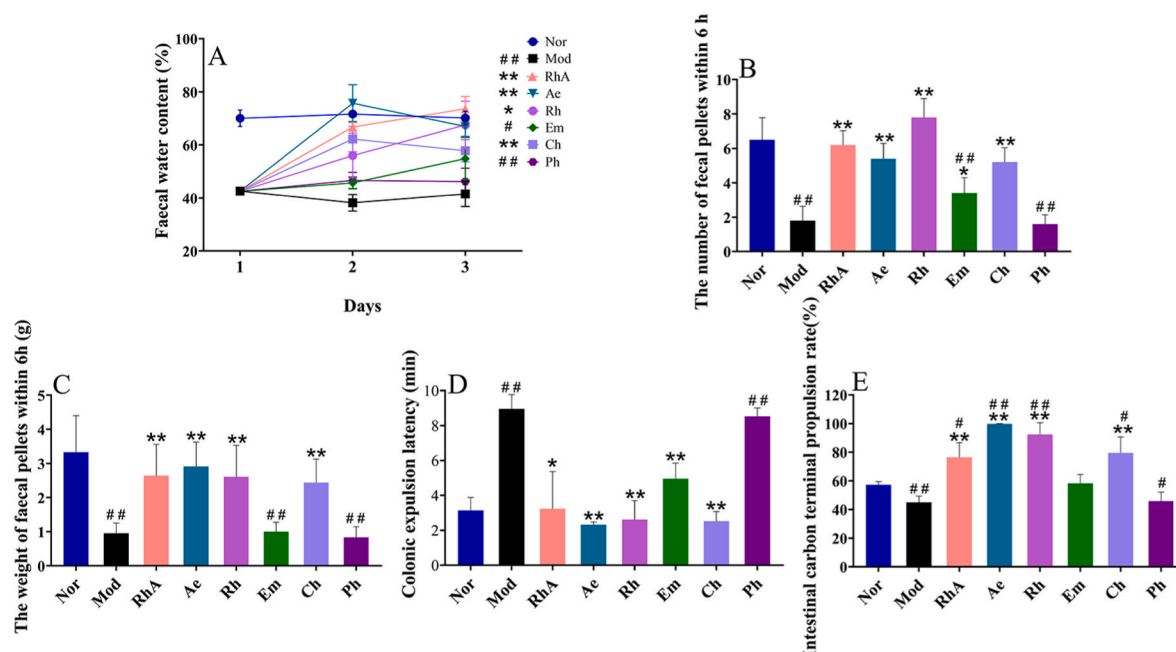


Fig. 4. Effect of RhA and its monomers on stool-softening. (A) Faecal water content; (B) number of faecal pellets; (C) weight of faecal pellets; (D) colonic expulsion latency; (E) small intestinal propulsion rate. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ vs. Nor group; $^*P < 0.05$, $^{**}P < 0.01$ vs. Mod group.

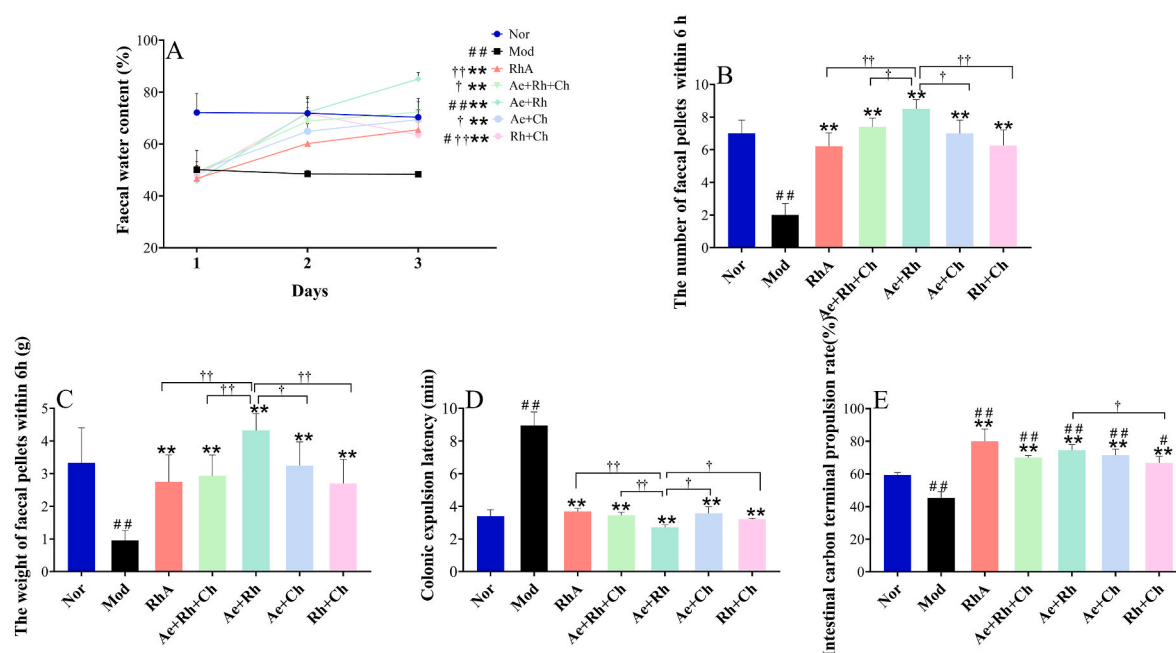


Fig. 5. Synergistic laxative effect of RhA and its monomers. (A) Faecal water content; (B) number of faecal pellets; (C) weight of faecal pellets; (D) colonic expulsion latency; (E) small intestinal propulsion rate. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ vs. Nor group; $^*P < 0.05$, $^{**}P < 0.01$ vs. Mod group; $^{\dagger}P < 0.05$, $^{\dagger\dagger}P < 0.01$ vs. Ae + Rh group.

3.4. The core targets and the binding energy

3.4.1. The core targets of rhubarb against constipation

Based on network pharmacology analysis, 992 intersection potential targets were obtained, and 103 core targets were particularly concerned, in which VIP, PKA, CFTR, NHE3, ENaC, AQP3, AQP8 were included (Fig. 7). In combination of the literature (Ikarashi et al., 2017), we rendered a hypothesis that the mechanism of rhubarb against constipation may be related to salt metabolism. KEGG analysis showed 15 main signal pathways, which involved PI3K-Akt, MAPK and cAMP signaling pathway and so on. In GO enrichment, the 1067 items were

involved, and the top 10 enriched biological processes, cellular components, molecular functions are presented as Fig. S8.

3.4.2. Molecular docking

As shown in Table 2, we performed molecular docking analysis on the core 7 key targets and the 5 bioactive components. The molecular docking results are shown in Fig. S9. The binding energy of VIP, PKA, CFTR, ENaC with all RhA monomers was lower than -5.0 kcal/mol. Additionally the binding energy of NHE3 with Rh, Ch, Ph was less than -5.0 kcal/mol, and the binding energy of AQP3 with Em, Ch was less than -5.0 kcal/mol. In contrast, the binding energy of AQP8 and all RhA

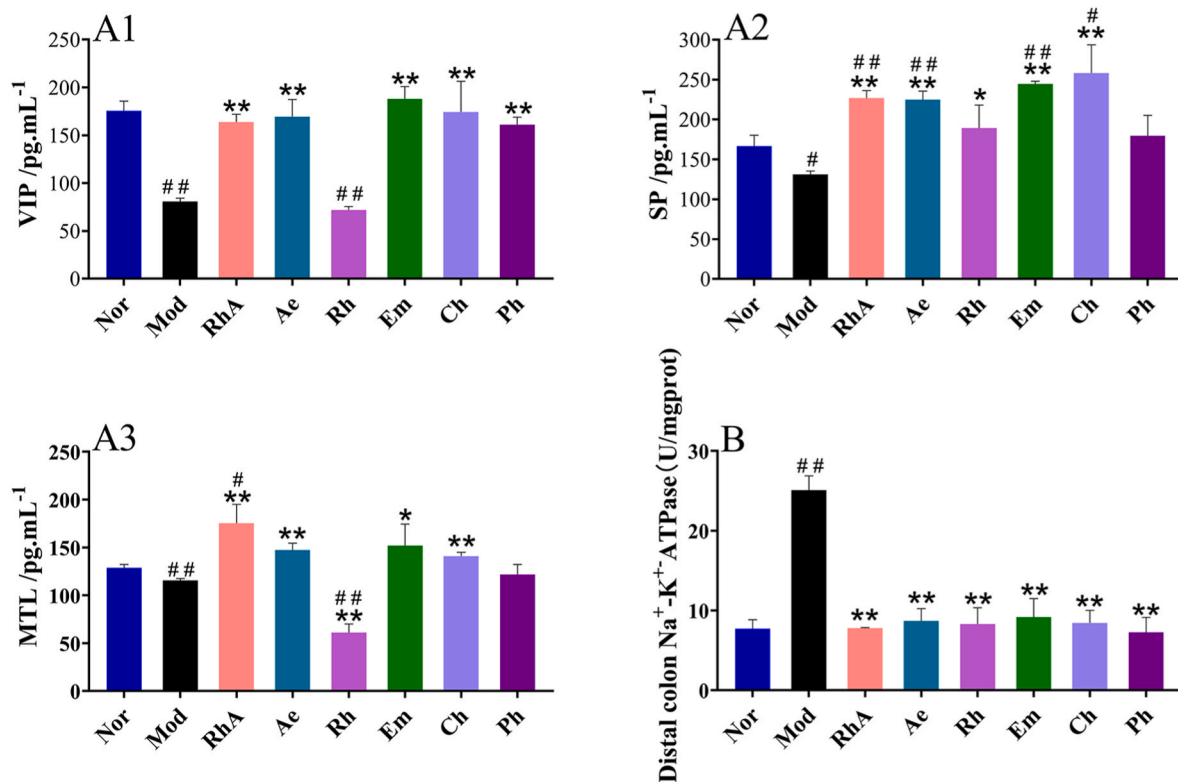


Fig. 6. Levels of VIP (A1), SP (A2), and MTL (A3) in rat serum and Na⁺-K⁺-ATPase (B) in colon tissues. [#]*P* < 0.05, ^{##}*P* < 0.01 vs. Nor group; ^{*}*P* < 0.05, ^{**}*P* < 0.01 vs. Mod group.

monomers was greater than -5.0 kcal/mol.

3.5. Histopathology and immunohistochemistry

The epithelial cells in the mucosal layer of colon tissues were intact, and the intestinal loops were arranged closely and neatly with abundant goblet cells in the Nor group. However, there are vacuoles filled with basophilic material in the Nor group but they are not observed in the Mod group. In contrast, the villus length of the distal colon in the Mod group was shortened (yellow arrow), and the colonic mucosa was thinner (black arrow). After intervention, the villus length recovered in comparison with the Mod group. The mucosal thickness in all treatment groups returned to the normal level. Additionally, where the red arrow points is consistent with leucocyte infiltration in the Em group, indicating that pathological damage might occur or, perhaps it could be lymphatic tissue (Fig. 8).

The expression levels of NHE3 and ENaC (denoted by the dark brown coarse-grained area in the Mod group) were significantly increased, while they were downregulated after treatment with RhA, Ae, Rh, and Ch (*P* < 0.05) (Fig. 9A and B).

The levels of AQP3 and AQP8 (denoted by the dark brown coarse-grained area in the Mod group) were significantly reduced in comparison with the Nor group, while they were upregulated after treatment with RhA, Ae, Rh, and Ch (*P* < 0.05) (Fig. 10A and B). However, the results actually demonstrated that Ph increased the expression level of AQP3 but not influenced that of AQP8, and Em increased the expression level of AQP8 but not influenced that of AQP3 as compared to the Mod group (*P* < 0.05).

3.6. Effect of RhA and its monomers on mRNA expression

As illustrated in Fig. 11, the mRNA expression levels of VIP, CAP1, PKA, CFTR, AQP3, AQP4, and AQP8 in the distal colon of the Mod group were lower than those of the Nor group, while the NHE3 and ENaC

expression levels were higher than those of the Nor group. RhA, Ae, and Rh significantly upregulated the mRNA expression of VIP, CAP1, PKA, CFTR, AQP3, AQP4, and AQP8 (*P* < 0.05); Ch significantly increased the mRNA expression of VIP, CAP1, PKA, CFTR, AQP3, and AQP4, but RhA, Ae Rh, and Ch downregulated the expression of NHE3 and ENaC in the constipated rat colonic tissues compared with the Mod group (*P* < 0.01). By contrast, Em did not significantly change the mRNA expression of VIP, PKA, CFTR, ENaC, and AQP8, and Ph did not significantly regulate the mRNA expression of VIP, CAP1, PKA, CFTR, NHE3, ENaC, and AQP8 compared with the Mod group (*P* > 0.05).

3.7. Effect of RhA and its monomers on protein expression

As shown in Fig. 12 and Suppl. Fig. S9 and Tables S2-S10, the expression levels of VIP, CAP1, PKA, CFTR, AQP3, AQP4, and AQP8 in the distal colon of the Mod group were decreased compared with those of the Nor group (*P* < 0.05). After treatment, RhA and Ae significantly upregulated the expression of all the above-mentioned proteins; Rh significantly increased the expression of the above-mentioned proteins, except VIP; Ch significantly increased the expression of the above-mentioned proteins, except AQP4, in the distal colon of the constipated rats (*P* < 0.05). Meanwhile, NHE3 and ENaC expression in the Mod group was very significantly higher than those of the Nor group, but they were significantly decreased after treatment with RhA, Ae, Rh, Em, and Ch (*P* < 0.05). By contrast, Em did not significantly change the expression of CAP1 and PKA, and Ph did not significantly regulate the expression of VIP, CAP1, PKA, CFTR, NHE3, ENaC, AQP3, AQP4, and AQP8 compared with the Mod group (*P* > 0.05).

3.8. Levels of osmolality and electrolyte ions in colonic fluid

As shown in Fig. 13A, compared with Nor group, the osmolality level in Mod group was lower (*P* < 0.05). Compared with Mod group, the osmolality of other treatment groups was significantly increased (*P* <

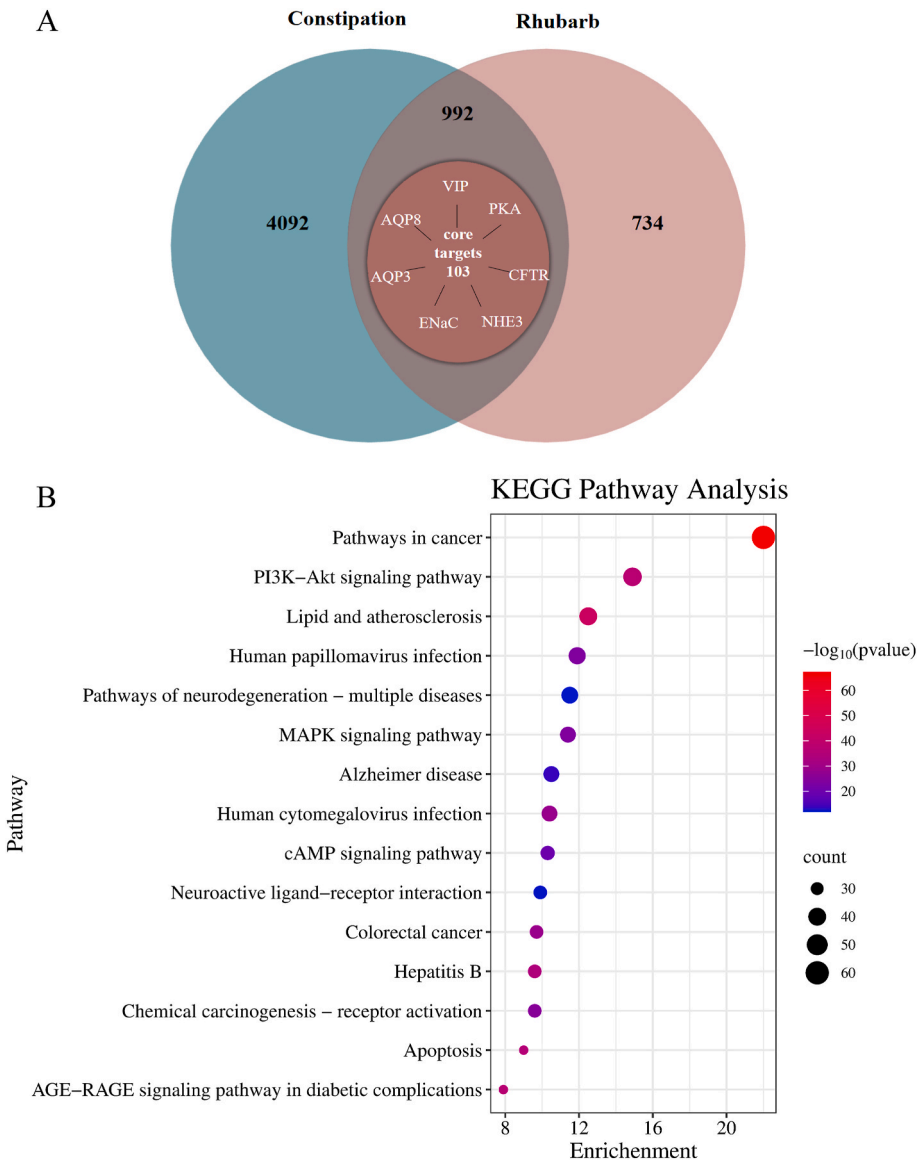


Fig. 7. Results of network pharmacology. (A) Venn diagram of the targets of rhubarb and constipation; (B) KEGG pathway enrichment.

Table 2
The binding energy of RhA monomers and key targets (kcal/mol).

Active compounds	Affinity (kcal/mol)						
	VIP	PKA	CFTR	NHE3	ENaC	AQP3	AQP8
Em	−5.57	−7.76	−6.92	−4.83	−5.99	−5.4	−4.38
Rh	−7.45	−6.38	−6.14	−6.1	−5.43	−4.76	−4.8
Ae	−5.52	−5.5	−6.29	−4.78	−5.43	−4.71	−4.77
Ch	−5.18	−7.51	−6.71	−5.15	−6.08	−5.54	−4.63
Ph	−5.69	−6.09	−6.09	−5.14	−6.39	−3.76	−4.56

0.05) except Ph treatment group. As shown in Fig. 13B and C, the concentrations of sodium and chloride ions in Mod group were significantly decreased compared with Nor group ($P < 0.05$). Compared with Mod group, the concentrations of Na^+ and Cl^- were significantly increased after treatment of RhA, Ae and Rh, while only Cl^- concentration increased significantly after treatment of Ch ($P < 0.05$). There was no significant difference in osmolality in Em and Ph treatment groups.

4. Discussion

The constipation model in mice was induced by forbidding water intake but allowing free access to food for 5 days according to the method described in the book Experimental Methodology of Pharmacology (Wei et al., 2010). However, the mice became severely malnourished with reduced food intake, weight loss, and even death between the third day and the fifth day. Some mice suffered tail loss and mania, and some were found to have had colon rupture after dissection. Considering that the tolerance of rats was greater than that of mice, male

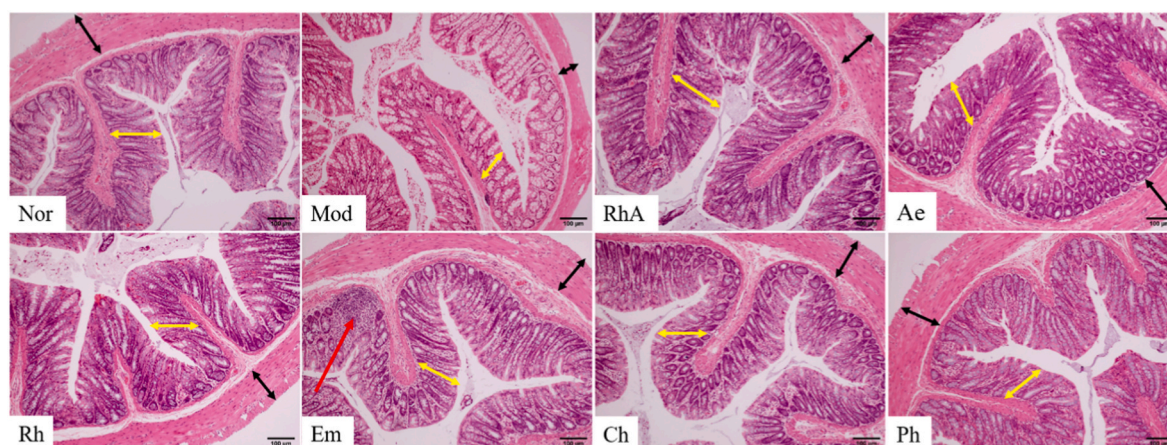


Fig. 8. Morphology of the colon stained with H&E in the constipated rats (original magnification of $100\times$ and scale bars = $100\mu\text{m}$). The yellow arrows indicate the length of the villus, and the black arrows indicate the thickness of muscles. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

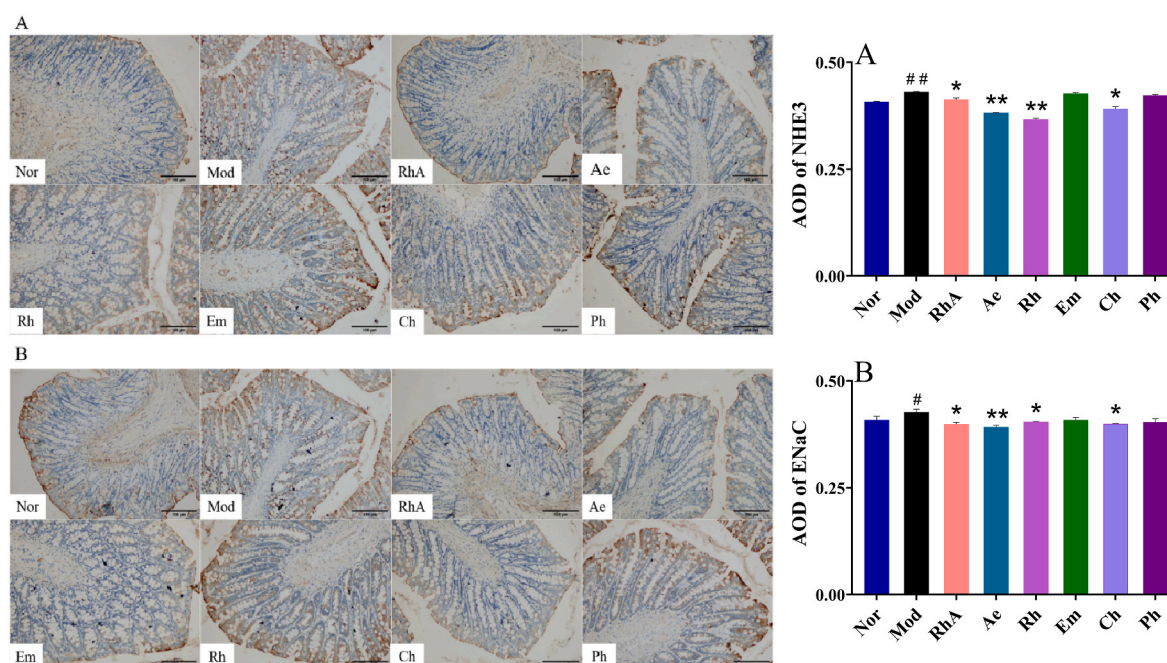


Fig. 9. Expression levels of NHE3 and ENaC in the colon (immunohistochemistry, original magnification of $200\times$ and scale bars = $100\mu\text{m}$). Expression levels of NHE3 in distal colon (A) expression levels of ENaC (B). $^{\#}P < 0.05$, $^{##}P < 0.01$ vs. Nor group; $^*P < 0.05$, $^{**}P < 0.01$ vs. Mod group.

SD rats were selected as experimental animals, and the same constipation-induced method was applied to rats. On the third day of modeling, the body weight of rats decreased by about 50 g, the stool became bead shaped, and the water content of faecal pellets decreased by 40–50%, at which point the RhA sample was administered. The body weight decreased after 3 days of administration, and the rats became severely malnourished. Therefore, we modified the method allowing the rats to ingest 3.0 mL of water by gavage twice a day for 5 successive days. The clinical observations of the model rats were described in Section 3.2.1, indicating that the rat constipation model was successfully established. Because the food had been chewed up and parts of it had fallen into the cage, we did not analyze correlation between food consumption and fecal mass. As to physical activity, we did observe some signs including hair shedding, activity frequency, weight loss, faecal water content, the number and weight of faecal pellets within 6 h and so on, but it was difficult to find quantification methods to determine all

physical signs.

In the pre-experiment, when the dose of RhA was set at 300 mg/kg, the water content, number, and weight of faecal pellets over 6 h on the third day of treatment and the colonic bead expulsion time of the model rats were all similar to those of normal rats. Therefore, 300 mg/kg RhA was selected as the medium dose in the experiment. The low dose (100 mg/kg) and high dose (500 mg/kg) were set to investigate dose-dependent relationship. To keep the total dose consistent with that of RhA, the total dose of RhA monomers or their combinations was set at 300 mg/kg.

The number, weight, and water content of faecal pellets were the most visible indicators for evaluating constipation in preclinical trials (Rasmussen and Pedersen, 2009). Our research showed that the water content, the pellet number, and the weight of faecal pellets over 6 h were significantly reduced in the water-deficit experimental rats from the third day on (Fig. 2), indicating that these rats had developed

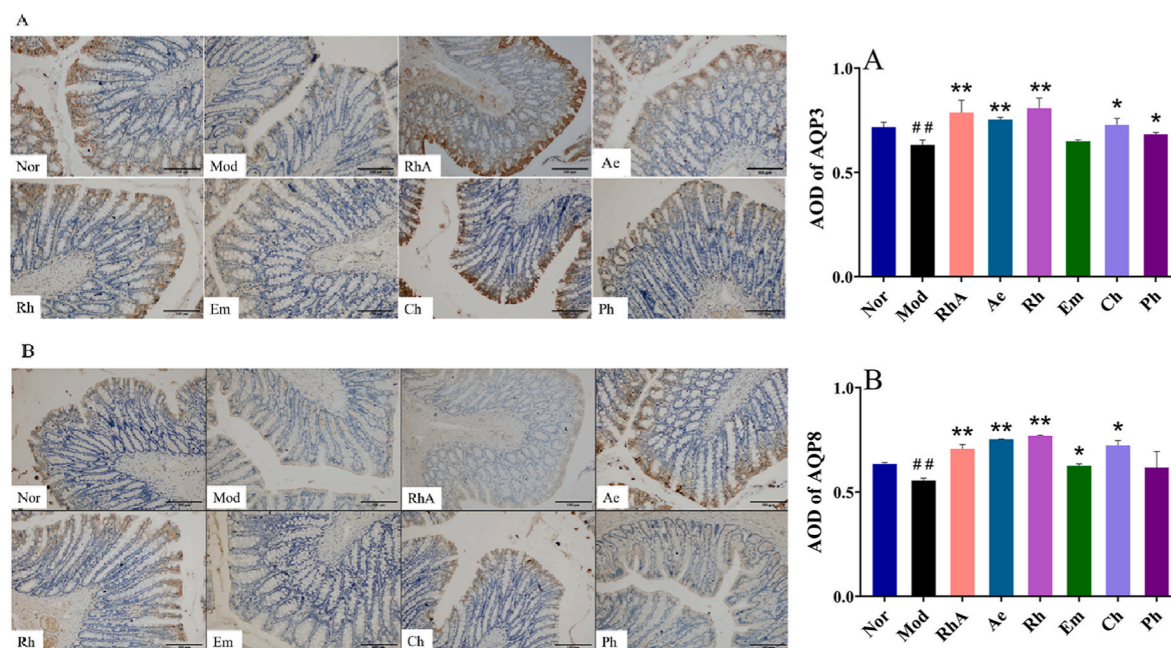


Fig. 10. Expression levels of AQP3 and AQP8 in the colon (immunohistochemistry, original magnification of 200 × and scale bars = 100 μm). Expression levels of AQP3 in distal colon (A) expression levels of AQP8 (B). * $P < 0.05$, ** $P < 0.01$ vs. Nor group; * $P < 0.05$, ** $P < 0.01$ vs. Mod group.

constipation. MRhA, HRhA, Ae, Rh, and Ch significantly increased the water content and the number and weight of faecal pellets within 6 h, shortened the colonic bead expulsion rate, and improved the small intestinal propulsion rate of the constipated rats (Figs. 3 and 4), suggesting that RhA, Ae, Rh, and Ch can effectively improve constipation. Additionally, all evaluation indicators of the LRhA group versus MRhA or HRhA group demonstrated a dose-dependent manner (Fig. 3). Compared to the Ae + Rh + Ch group after 3 days of experimental drug intervention, the faecal water content was significantly decreased in the Ch group; the 6-h defaecation number was significantly decreased in the Ae or Ch group; the colon expulsion latency was significantly shorter than that of the other monomer groups; the small-intestinal propulsion rate was significantly higher than that of the Ae or Ch group (Suppl. Fig. S4). Compared to the Ae + Rh group, the faecal water content was significantly decreased in the other monomer groups; the 6-h defaecation number was significantly decreased in the Ae group; the weight of faecal pellets was significantly decreased in the Ae or Rh group; the small-intestinal propulsion rate was significantly higher than that of the Ae or Rh group (Suppl. Fig. S5). Compared to the Ae + Ch group, the faecal water content was significantly decreased in the Ch group, the 6-h defaecation number was significantly decreased in the other monomer groups, the colon expulsion latency was significantly shorter than that of the Ae group, the small-intestinal propulsion rate was significantly higher than that of the Ae or Rh group (Suppl. Fig. S6). Compared to the Rh + Ch group, the small-intestinal propulsion rate was significantly higher in the Rh group (Suppl. Fig. S7). The above results suggested there were certain drug-interactions between the components. As a whole, the different combinations of Ae, Rh, and Ch exhibited a definite stool-softening effect, and the combination of Ae and Rh showed the strongest effect (Fig. 5), which may be related to the metabolism of sennosides as powerful laxatives. Recent studies have demonstrated that sennosides A, B, C, and D are the principal laxative components of rhubarb. Sennosides A and B were metabolised to rhein, and sennosides C and D were metabolised to Ae and Rh (Huang et al., 2019; Takayama et al., 2012; Wu et al., 2020). RhA and its monomers, including Ae, Rh, and Ch, softened the faecal pellets but did not lead to diarrhea as sennosides did, indicating that the laxative effect of Ae, Rh, and Ae is stronger than that of other RhA monomers but weaker than that of sennosides, which are basically consistent with those previously

reported (Liu et al., 2009; Yagi et al., 1997; Zhang et al., 2008), suggesting that Rh and/or Ae may be more suitable than their sennosides for the treatment of constipation.

It has been reported that the levels of neurotransmitter including serum VIP and SP, and the level of MTL mainly acted as gastrointestinal hormone, were altered in patients with constipation (Kitazawa and Kaiya, 2021). VIP and MTL can promote gastrointestinal motility (Geldre and Lefebvre, 2004; Tan et al., 2021). SP strongly promotes the contraction of smooth muscles and stimulates small intestinal and colonic mucosa to secrete water and electrolytes (Qian et al., 2015). In the study, the levels of serum VIP, MTL, and SP of the Mod group were significantly lower than those of the Nor group. However, serum VIP of the groups treated with RhA, Ae, Em, Ch, and Ph recovered to normal levels (Fig. 6A1). Serum SP levels of the RhA, Ae, Em, and Ch treatment groups were significantly increased (Fig. 6A2), and the serum MTL levels of the RhA, Ae, and Ch treatment groups were significantly improved compared with the Mod group (Fig. 6A3), while the serum MTL level of the Rh treatment group was significantly lower than that of the Mod group. Therefore, we speculated that RhA, Ae, and Ch may improve gastrointestinal motility by increasing serum VIP, SP, and MTL levels.

Colonic histopathologic features, such as the length of the villus layer and the thickness of the muscle layer, are considered important indices to evaluate the symptoms and clinical effect of constipation (Kim et al., 2017; Lee et al., 2012). The treatment with RhA and its monomers effectively protected the colonic histopathologic features with an increase in villus length and muscle layer thickness (Fig. 8). However, the Em showed some toxicity. The results suggested that Ae, Rh, Ch and Ph were the active components of RhA in the treatment of constipation.

In order to explore the signal pathway mechanism of rhubarb in the treatment of constipation, KEGG enrichment was performed based on the screening core targets of both rhubarb and constipation. As showed in Fig. 7, there were 15 signal pathways including PI3K-AKT, MAPK and cAMP signaling pathways. Among them, cAMP signaling pathway contained VIP, PKA, CFTR, NHE3, ENaC proteins, partly supported our hypothesis. Thus, cAMP signaling pathway was finally chosen as the focus to explore possible mechanism of action. The affinity between the receptor and the ligand is higher, and the conformation is more stable. It is generally accepted that binding energy less than −5 kcal/mol indicates a good binding activity (Li et al., 2021). The molecular docking

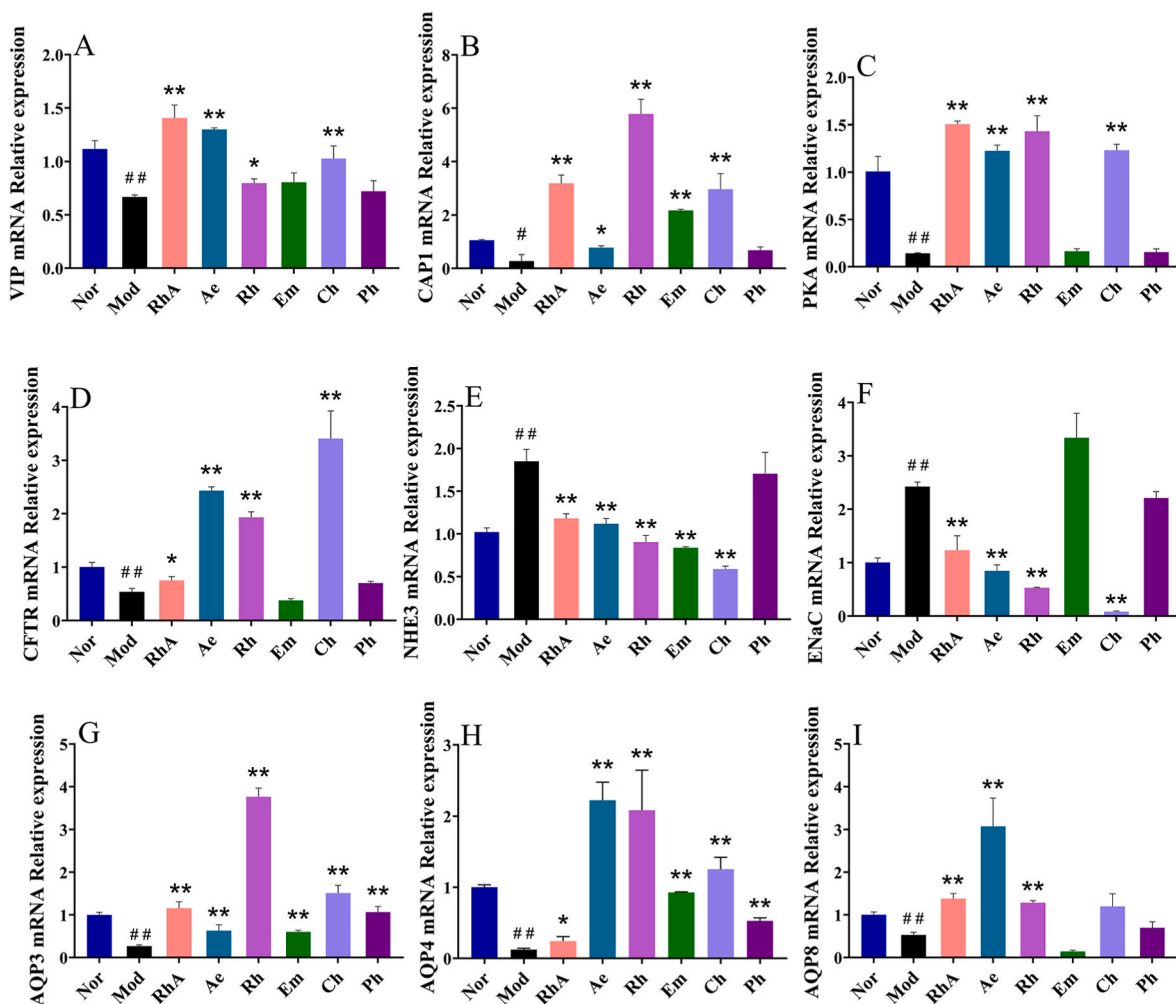


Fig. 11. VIP, CAP1, PKA, CFTR, NHE3, ENaC, AQP3, AQP4, and AQP8 mRNA expression in rat distal colon. #*P* < 0.05, ##*P* < 0.01 vs. Nor group; **P* < 0.05, ***P* < 0.01 vs. Mod group.

results show that these binding energies are all less than -5 kcal/mol (Table 2), indicating that the RhA monomers had high affinity with the core proteins of VIP, PKA, CFTR, ENaC, which further supported that cAMP signaling pathway possibly involved into stool-softening of rhubarb. CFTR is an anion channel expressed in the apical membrane of epithelial cells (Liu et al., 2017a; Moran, 2017), allowing the efflux of Cl^- and HCO_3^- ions into the enteric cavity (Bijvelds et al., 2015; Foulke-Abel et al., 2016). NHE3 and ENaC are expressed in the apical membrane of epithelial cells, allowing Na^+ and H_2O to remain in the enteric cavity (Waldman and Camilleri, 2018). CFTR inhibits the expression of ENaC, permitting colonic epithelial cells to limit the absorption of NaCl (Kunzelmann and Mall, 2002). The cAMP-PKA pathway activates CFTR and inhibits NHE3 channels (Foulke-Abel et al., 2016; Waldman and Camilleri, 2018). In the study, the expression of AQP3, AQP4, and AQP8 in the colonic tissue were downregulated in the Mod group, while they were upregulated in the RhA, Ae, Rh, and Ch treatment groups, indicating that the major mode for water to access colonic epithelial cells was mediated by the AQPs. The result was inconsistent with the previous report (Kon et al., 2014) that suggested a metabolite of sennoside A (rhein) triggered macrophage activation to decrease AQP3 expression in the colon, causing the laxative effect of rhubarb extract. The reason for the difference may be attributed to the research results at the cell are different from those obtained in animal. After the intervention with RhA, Ae, Rh, and Ch, the expression of VIP, CAP1, PKA, and CFTR was increased, while that of NHE3 and ENaC was decreased compared to the Mod group (Figs. 9–12). These results provided

evidence that VIP and the cAMP/PKA pathway play considerable roles in the biological effects of AQPs. Once RhA activated VIP expression, the cAMP/PKA pathway was stimulated. The cAMP/PKA further activated CFTR expression and inhibited NHE3 expression, and CFTR subsequently inhibited ENaC expression. The increase of CFTR and the decrease of NHE3 and ENaC expression led to much more Cl^- , HCO_3^- , and Na^+ in the enteric cavity, producing a hypertonic environment. This was supported by the improved osmolality level, Na^+ and Cl^- concentration in colonic fluid. The increased colonic fluid osmolality allowed water transit from the basolateral side to the colonic lumen, further increasing the expression of AQPs to adapt to the water concentration. As shown in Fig. 6B, the colonic concentration of Na^+/K^+ -ATPase was higher in the Mod group than in the Nor group, and it was lowered to normal levels after treatment with RhA and its monomers, which further evidenced that AQP expression was driven by the osmotic gradient. Overall, the expression of AQPs was regulated by VIP and the cAMP/PKA pathway, and cAMP/PKA further influenced the expression of CFTR, NHE3, and ENaC and/or the concentration of Na^+/K^+ -ATPase to produce a hypertonic state in the enteric cavity, accelerating water transport from the basolateral side to the colonic lumen. To adapt to water transport, AQP expression would be passively upregulated (Fig. 14).

5. Conclusions

A rat model of water deficit-induced constipation was successfully

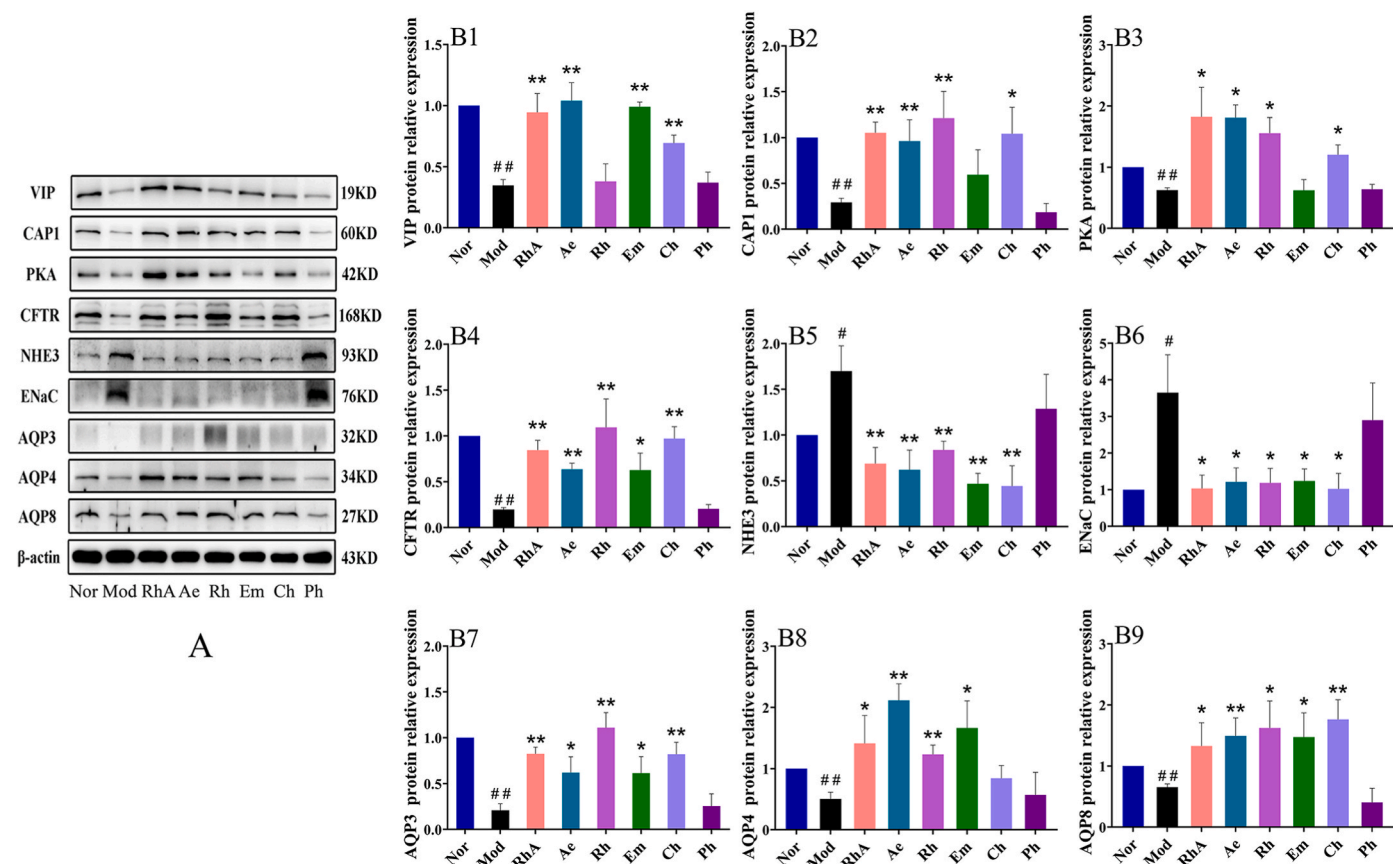


Fig. 12. Exposed protein bands (A) and grayscale ratios of VIP (B1), CAP1 (B2), PKA (B3), CFTR (B4), NHE3 (B5), ENaC (B6), AQP3 (B7), AQP4 (B8), and AQP8 (B9) protein expression in the colon of each group. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ vs. Nor group; $^*P < 0.05$, $^{**}P < 0.01$ vs. Mod group.

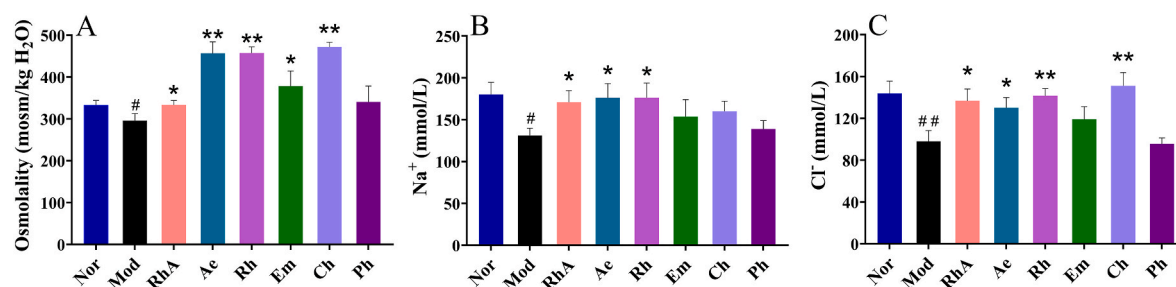


Fig. 13. Levels of osmolality(A) Na^+ and Cl^- in colonic fluid. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$ vs. Nor group; $^*P < 0.05$, $^{**}P < 0.01$ vs. Mod group.

established to imitate dry-stagnation type constipation. Aloe-emodin, rhein, and chrysophanol were the stool-softening components of rhubarb free anthraquinone extract, and there were certain drug-interactions between these components, which upregulated VIP expression, activated the cAMP/PKA pathway, stimulated CFTR expression, and inhibited NHE3 and ENaC expression, resulting in a hypertonic state in the colonic lumen. Water transport from the basolateral side to the enteric cavity could then be driven by an osmotic gradient, which in turn could lead to the upregulation of AQP3, AQP4, and AQP8 expression. In addition, rhubarb free anthraquinones improved gastrointestinal motility by increasing serum VIP, SP, and MTL levels to promote faecal excretion. Taken together, aloe-emodin, rhein and chrysophanol or their combinations have the potential to be developed into a biological agent to treat constipation.

CRedit authorship contribution statement

Huijuan Lv: Formal analysis; Original draft. Jingjing Niu: Formal analysis; Visualization. Wenhao Pan: Resources. Yudong Wang: Methodology. Lifang Wang: Methodology. Meng Wang: Visualization. Yali Shi: Data curation. Guifang Zhang: Investigation. Bandar Al Hamyari: Investigation. Shaohua Wang: Data curation. Xuefeng Li: Data curation. Yanbin Shi: Conceptualization; Supervision.

Declaration of competing interest

We declare that there is no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

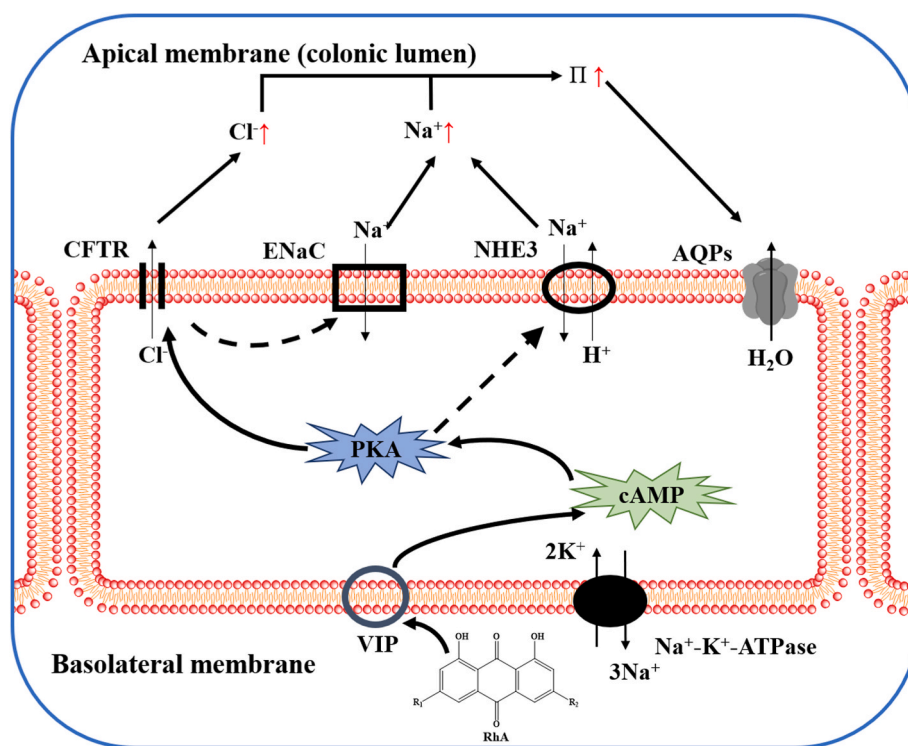


Fig. 14. Possible stool-softening mechanism of rhubarb free anthraquinones in rats.

Data availability

The data that has been used is confidential.

Acknowledgements

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.117336>.

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